

D425

Introduction to Chemistry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A hospital lab uses chlorine-37 as a tracer in a water-quality study that runs for six weeks. Chlorine has atomic number 17. The isotope is stored at 4 °C in amber bottles. How many neutrons does one atom of chlorine-37 contain?

- A. 20
- B. 37
- C. 54
- D. 17

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An ion has 12 protons and 10 electrons. What is its charge?

- A. +2
- B. +12
- C. 0
- D. -2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A materials team comparing sodium, magnesium, and chlorine (all in period 3) for a coating project notes that chlorine is a toxic gas and sodium reacts with water. Which of the three atoms has the largest atomic radius?

- A. Magnesium, because it sits in the middle
- B. Sodium
- C. Chlorine, because it has the most protons
- D. All three are the same size because they share period 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A student notes that carbon dioxide is produced by every cell in the body and makes up about 0.04% of air. Each C=O bond in the linear CO₂ molecule is polar. Is the CO₂ molecule polar or nonpolar, and why?

- A. Polar, because each C=O bond is polar.
- B. Polar, because it dissolves slightly in blood.
- C. Nonpolar, because it is a gas at room temperature.
- D. Nonpolar, because the linear shape cancels the two bond dipoles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A food scientist studying vinegar, which is about 5% acetic acid by volume and is sold in glass bottles, examines the acetic acid structure CH₃COOH. Which functional group gives acetic acid its acidic behavior?

- A. Methyl group (–CH₃)
- B. Alcohol (–OH)
- C. Carboxylic acid (–COOH)
- D. Amine (–NH₂)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Propane burns in a patio heater that a hotel runs from 6 p.m. to midnight. The skeleton equation is C₃H₈ + __O₂ → 3CO₂ + 4H₂O. The heater's tank holds 20 pounds of fuel. What coefficient goes in front of O₂?

- A. 10
- B. 3
- C. 5
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A student balances $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ by rewriting the product as H_2O_2 . What is wrong with this approach?

- A.** Nothing; any change that makes the atom counts match is allowed.
- B.** The error is that the coefficients must add to the same total on both sides.
- C.** The student should have changed the subscript on H_2 instead.
- D.** Changing a subscript creates a different substance (hydrogen peroxide), so the equation no longer describes the reaction.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

In a water-treatment lab that runs tests every morning at 7 a.m., a technician mixes silver nitrate solution with sodium chloride solution, and a white solid forms. The reagents cost about \$40 per bottle. How is this reaction classified?

- A.** Combination
- B.** Double replacement (precipitation)
- C.** Combustion
- D.** Single replacement

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A chemistry teacher who has taught for 20 years lists four substances: CH_4 , H_2S , Ne, and H_2O . She adds that H_2S smells like rotten eggs. Which has the highest boiling point?

- A.** Ne, because noble gases are the most stable
- B.** H_2O
- C.** CH_4 , because it has the most hydrogen atoms
- D.** H_2S , because it has the greatest molar mass

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Substance X boils at a much higher temperature than a nonpolar substance of similar molar mass. What best explains the difference?

- A.** X has stronger covalent bonds inside each molecule.
- B.** X has weaker London dispersion forces.
- C.** X must be an ionic compound.
- D.** X has stronger intermolecular forces, such as hydrogen bonding or dipole-dipole attraction.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A pharmacy stores a phosphate buffer, which a supplier ships in 500 mL bottles, at pH 7.4. A technician adds a few drops of dilute HCl. The bottle was opened last Tuesday. What happens to the pH?

- A.** It rises, because the buffer releases more acid.
- B.** It falls sharply to about 1, as it would in pure water.
- C.** It drops only slightly, because the buffer's base component neutralizes the added acid.
- D.** It stays exactly at 7.4.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Solution A has pH 3 and solution B has pH 5. How does the hydrogen ion concentration of A compare with B?

- A.** A has 100 times more H^+ than B.
- B.** They are the same, because both are acidic.
- C.** A has 2 times more H^+ than B.
- D.** A has 100 times less H^+ than B.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A hospital lab uses chlorine-37 as a tracer in a water-quality study ~~that runs for six weeks~~. Chlorine has atomic number 17. ~~The isotope is stored at 4 °C in amber bottles.~~ How many neutrons does one atom of chlorine-37 contain?

Target word: neutron count **Decides:** Neutrons = mass number 37 – atomic number 17.

A	Correct	$37 - 17 = 20$ neutrons.
B	Confusion set	That is the mass number: protons plus neutrons.
C	Inverted operation	Adds 37 and 17 instead of subtracting.
D	Wrong stem number	That is the proton count (atomic number).

2. Answer A Concept 1 • transfer

An ion has 12 protons and 10 electrons. What is its charge?

Target word: solve for charge instead of neutrons **Decides:** Charge = protons – electrons = $12 - 10$.

A	Correct	Two more protons than electrons gives a 2+ charge (Mg^{2+}).
B	Wrong stem number	The proton count, not the net charge.
C	Dropped element	Ignores the electron count; only equal protons and electrons are neutral.
D	Inverted operation	Subtracts in the wrong direction; losing electrons makes a positive ion.

3. Answer B Concept 2

A materials team comparing sodium, magnesium, and chlorine (all in period 3) ~~for a coating project notes that chlorine is a toxic gas and sodium reacts with water.~~ Which of the three atoms has the largest atomic radius?

Target word: largest radius in one period **Decides:** Radius shrinks left to right as nuclear charge rises in the same shell.

A	Dropped element	Position in the middle does not make it largest; the trend runs one way across the row.
B	Correct	It is farthest left in period 3, so it has the least nuclear pull on the same outer shell.
C	Inverted operation	More protons in the same shell pull electrons closer, making the atom smaller.
D	Absolute language	Sharing a shell count does not make radii equal; nuclear charge changes across the row.

4. Answer D Concept 3

A student notes that carbon dioxide ~~is produced by every cell in the body and makes up about 0.04% of air.~~ Each C=O bond in the linear CO₂ molecule is polar. Is the CO₂ molecule polar or nonpolar, and why?

Target word: polarity of the whole molecule **Decides:** Linear shape makes the two equal bond dipoles cancel.

A	Dropped element	Drops the shape; polar bonds alone do not make a polar molecule.
B	Camouflage	Leans on the body detail; slight solubility does not decide molecular polarity.
C	Adjacent category	State of matter does not set polarity; water vapor is a gas and is polar.
D	Correct	Polar bonds pointing in opposite directions cancel, leaving no net dipole.

5. Answer C Concept 4

A food scientist studying vinegar, ~~which is about 5% acetic acid by volume and is sold in glass bottles,~~ examines the acetic acid structure CH_3COOH . Which functional group gives acetic acid its acidic behavior?

Target word: functional group responsible **Decides:** The $-\text{COOH}$ group releases H^+ .

A	Adjacent category	An alkyl group is not reactive in this way; it is just part of the carbon chain.
B	Confusion set	The $-\text{OH}$ inside $-\text{COOH}$ acts as part of the carboxyl group; alcohols alone are not meaningfully acidic.
C	Correct	The $-\text{COOH}$ group donates a proton, which makes the compound acidic.
D	Inverted operation	Amines are basic, the opposite behavior.

6. Answer C Concept 5

Propane burns in a patio heater ~~that a hotel runs from 6 p.m. to midnight.~~ The skeleton equation is $\text{C}_3\text{H}_8 + __\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$. ~~The heater's tank holds 20 pounds of fuel.~~ What coefficient goes in front of O_2 ?

Target word: O_2 coefficient **Decides:** Products hold $6 + 4 = 10$ O atoms; $10 / 2 = 5$ O_2 molecules.

A	Confusion set	Counts oxygen atoms instead of O_2 molecules.
B	Dropped element	Counts only the oxygen in CO_2 (6 atoms) and drops the water.
C	Correct	3CO_2 has 6 O and $4\text{H}_2\text{O}$ has 4 O: 10 O atoms = 5 O_2 molecules.
D	Wrong stem number	The water coefficient, not the O_2 coefficient.

7. Answer D Concept 5 · transfer

A student balances $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ by rewriting the product as H_2O_2 . What is wrong with this approach?

Target word: reverse: identify the illegal balancing move **Decides:** Subscripts define the substance; only coefficients may change.

A	Absolute language	Only coefficient changes are allowed; subscripts are fixed by the formula.
B	Confusion set	Atoms must balance, not the sum of coefficients.
C	Adjacent category	Still a subscript change, which is not allowed.
D	Correct	Balancing must use coefficients: $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$.

8. Answer B Concept 6

In a water-treatment lab ~~that runs tests every morning at 7 a.m.,~~ a technician mixes silver nitrate solution with sodium chloride solution, and a white solid forms. ~~The reagents cost about \$40 per bottle.~~ How is this reaction classified?

Target word: reaction type **Decides:** Two ionic compounds swap partners: $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{AgCl} + \text{NaNO}_3$.

A	Confusion set	Two reactants do not make a combination; the products here are two new compounds, not one.
B	Correct	Cations and anions swap partners and insoluble AgCl precipitates.
C	Inverted operation	Combustion needs a fuel reacting with O_2 ; no oxygen gas is involved.
D	Adjacent category	Single replacement has one element displacing another; here both compounds exchange ions.

9. Answer B Concept 7

A chemistry teacher ~~who has taught for 20 years~~ lists four substances: CH₄, H₂S, Ne, and H₂O. ~~She adds that H₂S smells like rotten eggs.~~ Which has the highest boiling point?

Target word: highest boiling point **Decides:** Only water has hydrogen bonding (H on O).

A	Camouflage	Chemical stability does not decide boiling point; Ne has only weak London forces.
B	Correct	Hydrogen bonding makes water's intermolecular forces far stronger than the others'.
C	Adjacent category	CH ₄ is nonpolar; H on carbon does not form hydrogen bonds.
D	Confusion set	Mass matters for London forces, but H ₂ S cannot hydrogen-bond like water.

10. Answer D Concept 7 • transfer

Substance X boils at a much higher temperature than a nonpolar substance of similar molar mass. What best explains the difference?

Target word: reverse: from observed boiling point to cause **Decides:** Boiling separates molecules; it does not break covalent bonds.

A	Confusion set	Boiling does not break bonds inside molecules; it overcomes forces between them.
B	Inverted operation	Weaker forces would lower the boiling point, not raise it.
C	Absolute language	Polar molecular compounds can also boil high; nothing requires X to be ionic.
D	Correct	Stronger attraction between molecules takes more energy to overcome.

11. Answer C Concept 8

A pharmacy stores a phosphate buffer, ~~which a supplier ships in 500 mL bottles,~~ at pH 7.4. A technician adds a few drops of dilute HCl. ~~The bottle was opened last Tuesday.~~ What happens to the pH?

Target word: buffer response **Decides:** The buffer's conjugate base neutralizes the small amount of added acid.

A	Inverted operation	Added acid pushes pH down, not up.
B	Adjacent category	That describes unbuffered water, not a buffer.
C	Correct	Buffers resist but do not fully prevent pH change.
D	Absolute language	A buffer resists change; it does not hold pH perfectly fixed.

12. Answer A Concept 8 • transfer

Solution A has pH 3 and solution B has pH 5. How does the hydrogen ion concentration of A compare with B?

Target word: ratio of [H⁺] **Decides:** Each pH unit is a factor of 10: $10^{(5-3)} = 100$.

A	Correct	Two pH units = $10 \times 10 = 100$ -fold difference; lower pH means more H ⁺ .
B	Adjacent category	Both being below 7 does not make their concentrations equal.
C	Confusion set	Treats the log scale as linear.
D	Inverted operation	Lower pH means more H ⁺ , not less.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Atomic structure	Protons fix the element; neutrons = mass number – atomic number; charge = protons – electrons.	Treating mass number as the neutron count, or letting electrons change the element.	A mass number or an ion charge next to an atomic number.
2	Periodic trends	Atomic radius grows down a group and shrinks left to right; electronegativity and ionization energy do the reverse.	Assuming more protons always means a bigger atom.	Two or more elements from the same row or column compared on size or pull.
3	Molecular shape and polarity	A molecule is polar only if it has polar bonds AND a shape that does not cancel them.	Calling any molecule with polar bonds a polar molecule.	A symmetric shape (linear, tetrahedral) with polar bonds.
4	Functional groups	The functional group, not the carbon chain, sets how an organic molecule reacts: –OH alcohol, –COOH carboxylic acid, –NH ₂ amine, C=O carbonyl.	Naming a compound by chain length and ignoring the group that reacts.	A structure fragment such as –COOH or –OH in the stem.
5	Balancing equations	Balance with coefficients so every atom count matches on both sides; never change subscripts.	Balancing by changing a formula's subscripts, or counting atoms instead of molecules.	"Coefficient" or an unbalanced skeleton equation.
6	Reaction types and products	Classify by pattern: combination, decomposition, single replacement, double replacement, combustion; the pattern predicts the products.	Calling any reaction with two reactants a combination reaction.	Two ionic compounds swapping partners, or a fuel plus O ₂ .
7	Intermolecular forces	Boiling point tracks the forces between molecules: hydrogen bonding > dipole-dipole > London dispersion at similar size.	Explaining boiling point with the covalent bonds inside the molecule.	Compare boiling points, or H bonded to N, O, or F.
8	Acids, bases, and buffers	pH = –log[H ⁺]; each pH unit is a tenfold change; a buffer (weak acid + conjugate base) resists pH change.	Reading the pH scale as linear, or thinking a buffer keeps pH perfectly fixed.	Two pH values to compare, or a weak acid paired with its salt.

Course scope (public description): In Introduction to Chemistry, learners will discover the impact of chemistry on everyday life. They'll learn about the structure of the atom, study periodic trends, analyze the structure of molecules and their properties, and describe the importance of common functional groups within the periodic table. They'll identify balanced chemical equations, describe types of chemical reactions and predict their products, and examine intermolecular forces and describe their impact on the properties of substances. Finally, they'll study the properties of acids, bases, and buffer systems, and properties unique to liquids and gases.